

30 Days SQL Micro Course Assignment Answer

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```
DROP TABLE IF EXISTS hospital_data;
```

```
--CREATE TABLE hospital_data--
```

```
CREATE TABLE hospital_data(  
    Hospital_Name VARCHAR(100),  
    hospital_Location VARCHAR(100),  
    Department VARCHAR(100),  
    Doctors_Count INT,  
    Patients_Count INT,  
    Admission_Date VARCHAR(20),  
    Discharge_Date VARCHAR(20),  
    Medical_Expenses NUMERIC(10,2)  
);
```

```
SELECT * FROM hospital_data;
```

1. Total Number of Patients

```
SELECT SUM(Patients_Count) AS total_patients  
FROM hospital_data;
```

2. Average Number of Doctors per Hospital.

```
SELECT Hospital_Name, AVG(Doctors_Count) AS avg_doctors  
FROM hospital_data  
GROUP BY Hospital_Name;
```

3. Top 3 Departments with the Highest Number of Patient.

```
SELECT Department, SUM(Patients_Count) AS total_patients  
FROM hospital_data  
GROUP BY Department  
ORDER BY total_patients DESC  
LIMIT 3;
```

4. Hospital with the Maximum Medical Expenses.

```
SELECT Hospital_Name, SUM(Medical_Expenses) AS total_expenses
FROM hospital_data
GROUP BY Hospital_Name
ORDER BY total_expenses DESC
LIMIT 1;
```

5. Daily Average Medical Expenses.

```
SELECT Hospital_Name , Admission_Date, AVG(Medical_Expenses) AS avg_expenses
FROM hospital_data
GROUP BY Hospital_Name, Admission_Date
ORDER BY Admission_Date;
```

6. Longest Hospital Stay.

```
SELECT Hospital_Name,
Department,
Admission_Date,
Discharge_Date,
(TO_DATE(Discharge_Date,'DD-MM-YYYY')
-
TO_DATE(Admission_Date,'DD-MM-YYYY')) AS stay_days
FROM hospital_data
ORDER BY stay_days DESC
LIMIT 1;
```

7. Total Patients Treated Per City.

```
SELECT Hospital_Location , SUM(Patients_Count) AS total_patients
FROM hospital_data
GROUP BY Hospital_Location
ORDER BY total_patients DESC;
```

8. Average Length of Stay Per Department.

```
SELECT Department,AVG(TO_DATE(Discharge_Date,'DD-MM-YYYY')-TO_DATE(Admission_Date,'DD-MM-YYYY'))
AS avg_stay
FROM hospital_data
GROUP BY Department;
```

9. Identify the Department with the Lowest Number of Patients.

```
SELECT Department , SUM(Patients_Count) AS total_patients
FROM hospital_data
GROUP BY Department
ORDER BY total_patients
LIMIT 1;
```

10. Monthly Medical Expenses Report.

```
SELECT
TO_CHAR(TO_DATE(Admission_Date,'DD-MM-YYYY'),'YYYY-MM') AS month,
SUM(Medical_Expenses) AS total_expenses
FROM hospital_data
GROUP BY month
ORDER BY month;
```
